RES-DAT — Environmental Data and Case-Study Definition

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1 Domain Purpose and Scope

RES-DAT defines the environmental data and case-study evidence needed to reconstruct one selected historical tsunami event, validate that event against authoritative observations, and support documented defence-specific impact cases. The initial project retains one primary historical event and controlled physical variations around that event. It does not add unrelated tsunami events or build a broad multi-event ML dataset during the initial scope.

The active studentship case is the 2011 Tōhoku event with two regional-to-local corridor trajectories: epicentre to the Kamaishi region and epicentre to the Sendai coastal plain. The active use of the data is impact modelling, Tōhoku/local comparison and barrier-class comparison. ML datasets, surrogate optimisation and generative geometry remain future work until the event reconstruction, local hydrodynamic load extraction and validation hierarchy are stable.

The Domain shall also catalogue defence geometry and local exposure evidence where available, but damage, material deformation, scour and FSI remain stretch extensions after hydrodynamic load extraction is stable. Source role: project-scope baseline and data-interface decision. Supporting sources: event metadata from U.S. Geological Survey [1], onshore observations from Mori, Takahashi, and The 2011 Tohoku Earthquake Tsunami Joint Survey Group [2], offshore observations from National Oceanic and Atmospheric Administration [3] and NOAA National Centers for Environmental Information [4], and terrain source candidates in Japan Oceanographic Data Center [5], GEBCO Bathymetric Compilation Group 2026 [6], NOAA National Centers for Environmental Information [7] and Geospatial Information Authority of Japan [8]. Limitation: the cited source records support data selection and handoff requirements; they do not demonstrate that the project has executed validation.

2 Domain Deliverables

3 RES-DAT-GEO — Bathymetry, Topography and Coastal Geometry

3.1 Purpose and Scope

This Deliverable defines the bathymetry, topography and coastal-geometry data required to build the regional 2D NLSWE domain, the two regional-to-local corridor extracts and the local 3D URANS–VOF terrain surfaces. It covers dataset selection, fallback hierarchy, coordinate reference systems, vertical datums, preprocessing and handoff requirements. It does not define arbitrary corridor sketches: the corridor footprint is a data/model interface constructed from the event source, target coast, observation footprint, defence geometry and numerical-boundary requirements.

3.2 Research Questions

The data decision asks:

1. Which topobathymetric source is authoritative inside the Japan nearshore and local corridors?
2. Which global products are acceptable for the outer regional domain or fallback coverage?
3. How are horizontal CRS, vertical datum, sign convention, resolution and licence metadata preserved through clipping and interpolation?
4. Which preprocessing uncertainty must be handed to ‘RES-VER‘ and Software before validation claims are made?

3.3 Terminology and Definitions

3.4 Literature Evidence

Japan Oceanographic Data Center [5] documents J-EGG500 as a 500 m gridded bathymetric dataset around Japan compiled from depth measurements from Japanese marine survey institutions. Source role: supports the primary Japan-region bathymetry candidate. Project-specific conclusion: J-EGG500 is the preferred first candidate for offshore bathymetry in the Kamaishi and Sendai corridors if licence, datum and import metadata are acceptable. Limitation: the source page notes heterogeneous data quality and smoothing, so local relief and harbour/breakwater details shall not be assumed resolved.

GEBCO Bathymetric Compilation Group 2026 [6] provides the current global 15 arc-second terrain grid for ocean and land and a Type Identifier grid. Source role: supports outer-domain and fallback topobathymetry. Project- specific conclusion: GEBCO is selected as the first global fallback and outer-domain source, not as the preferred source for local defence geometry. Limitation: GEBCO’s vertical datum is treated as mean sea level while shallow-water source datums can vary, so nearshore datum reconciliation remains mandatory.

NOAA National Centers for Environmental Information [7] provides NOAA ETOPO 2022 as a 15 arc-second global relief model designed for applications including tsunami forecasting and modelling. Source role: supports secondary global fallback and cross-check. Project-specific conclusion: ETOPO shall be used to fill or audit outer-domain coverage if J-EGG500/GEBCO coverage or provenance is insufficient. Limitation: it is not selected for local barrier-scale topography.

Geospatial Information Authority of Japan [8] documents GSI digital elevation models with 1 m, 5 m and 10 m mesh products. Source role: supports Japan onshore topography. Project-specific conclusion: GSI DEM is the preferred onshore topographic source for corridor clipping and inundation/run-up comparison where available. Limitation: mixed DEM classes have different survey methods and accuracy; class boundaries shall be retained in the provenance manifest.

3.5 Governing Relations and Quantitative Definitions

3.6 Required Data Variables and Units

3.7 Candidate Data Sources

The selected/fallback hierarchy for the Tōhoku case is:

1. **Japan offshore bathymetry:** J-EGG500 [5], subject to licence, datum and coverage checks for the selected corridors.
2. **Japan onshore topography:** GSI DEM [8], selecting the finest available mesh class that is legally usable and spatially continuous over each corridor.
3. **Outer regional and fallback topobathymetry:** GEBCO_2026 [6].
4. **Secondary global fallback or cross-check:** NOAA ETOPO 2022 [7].

The selected approach is a stitched and provenance-preserving terrain model rather than one unqualified raster. Each cell or clipped raster tile shall retain the source product, source resolution, interpolation method, horizontal CRS, vertical datum and sign convention.

3.8 Spatial and Temporal Coverage

Coverage shall include:

- the outer Tōhoku regional propagation domain, including artificial-boundary and sponge regions;
- the epicentre-to-Kamaishi corridor and enough lateral margin to contain the severe-impact observations, bay/coastline geometry and relaxation zones;
- the epicentre-to-Sendai coastal-plain corridor and enough inland coverage to include observed inundation limits;
- local 3D terrain patches around wall-type and obstacle/dissipating barrier classes.

Topobathymetric products are treated as static during one hydrodynamic run. The earthquake source updates the bed through ‘RES-DAT-SRC‘; it does not change the static terrain-data selection.

3.9 Coordinate and Datum Requirements

The regional model uses a horizontal projected coordinate frame (x, y) and a positive-upward bed elevation b. Source datasets shall record horizontal CRS, vertical datum, units, coastline convention and any tidal or geoid correction used before interpolation to the finite-volume mesh.

Base acquisition coordinates shall remain longitude/latitude (λ, ϕ) until the corridor target points are finalised in GIS. The solver-facing terrain shall be delivered in a projected metric CRS chosen for the final regional and local domains. Any dataset whose horizontal CRS or vertical datum cannot be confirmed shall be marked "usable only for visual screening" until metadata are resolved.

3.10 Data Processing and Transformation

The required transformation pipeline is:

$$(\lambda, \phi, z) \to (E, N, b_0) \to (x, y, b_0) \to b_0^{\text{mesh}} \to b_{0,i} \to b_i(t) \tag{3.1}$$

For positive-upward elevation data:

$$b_0 = z - z_{\text{ref}} \tag{3.2}$$

For raw depth d positive downward:

$$b_0 = -d - z_{\text{ref}} \tag{3.3}$$

The cell-average static bed is:

$$b_{0,i} = \frac{1}{A_i} \int_{K_i} b_0^{\text{mesh}}(x, y)\, \mathrm{d}A \tag{3.4}$$

QGIS/geospatial work owns dataset selection, domain placement, coordinate transformation and source-data preparation. It is a dependency of the regional model, not part of the mathematical model itself.

3.11 Uncertainty, Provenance and Licensing

The terrain manifest shall store, for every source tile or derived terrain surface:

- source product, provider, citation key and access URL;
- licence or usage note;
- horizontal CRS, vertical datum and sign convention;
- native resolution and interpolation or smoothing operations;
- clipping polygon, corridor identifier and mesh-projection method;

- uncertainty flags for datum conversion, coastline alignment, resampling, smoothing and source-age heterogeneity.

Project conclusion: a terrain raster without this metadata is not an accepted validation input. Limitation: the metadata requirements do not quantify the full bathymetric error; they make the uncertainty auditable by ‘RES-VER‘.

3.12 Data Acceptance Criteria

3.13 Candidate Approaches

3.14 Critical Comparison

3.15 Assumptions and Applicability

3.16 Limitations and Failure Conditions

3.17 Project-Specific Conclusions

The Tōhoku case-study terrain workflow shall use Japan-specific sources where available and global products only where they are physically and numerically appropriate. J-EGG500 and GSI DEM are selected as the first candidates for Japan offshore and onshore data respectively. GEBCO_2026 is selected for outer-domain/fallback coverage, with ETOPO 2022 retained as a secondary fallback and consistency check. The local barrier mesh shall not infer defence geometry from 15 arc-second global products.

3.18 Selected Approach and Rationale

Use a provenance-preserving terrain hierarchy:

> J-EGG500 offshore bathymetry + GSI onshore DEM where available, merged with GEBCO_2026 for outer-domain and fallback coverage, and ETOPO 2022 as a secondary fallback/cross-check

Source role: supports terrain-data finalisation for the regional model, the two corridor extracts and local 3D terrain. Supporting citations: Japan Oceanographic Data Center [5], Geospatial Information Authority of Japan [8], GEBCO Bathymetric Compilation Group 2026 [6] and NOAA National Centers for Environmental Information [7]. Downstream handoff: Software shall ingest these sources through a dataset manifest, perform coordinate transformation, clip corridor rasters, interpolate to mesh/field containers and write structured outputs with source provenance.

3.19 Unresolved Decisions

The mathematical data requirements are established. The actual QGIS domain lock, final bathymetry/topograph dataset, datum reconciliation, coastline treatment, smoothing policy and mesh-projection workflow remain open.

Open source-specific decisions are the final J-EGG500 and GSI licence acceptance, any required Japanese geodetic/vertical datum conversion, GEBCO/ETOPO blending boundary, and whether higher-resolution site-specific harbour or defence surveys can be obtained for Kamaishi. TODO–missing-source: no dedicated local Kamaishi bay-mouth breakwater geometry or survey source has yet been confirmed in the WBS bibliography.

3.20 Requirements and Handoffs

The data workstream shall provide mesh-projected static bed values, source-bed increments where applicable, domain boundaries, sponge-layer candidate regions and provenance metadata before validation or implementation can be declared complete.

Software-facing requirements now begin in parallel with remaining research: dataset manifest, coordinate-transform workflow, corridor-polygon generation, raster/vector clipping, bathymetry/topograpl interpolation, HDF5 or equivalent structured terrain output, and replay-boundary metadata for the local 3D inlet.

3.21 Deliverable Completion Record

4 RES-DAT-SRC — Earthquake and Tsunami Source Data

4.1 Purpose and Scope

This Deliverable defines the earthquake event metadata, finite-fault source candidates and bed-deformation data required to reconstruct the 2011 Tōhoku tsunami in the regional 2D NLSWE model. It covers source data selection and projection into model-ready bed increments. It does not choose a structural damage model, a local barrier geometry or a solver implementation.

4.2 Research Questions

The source-data decision asks:

1. Which event metadata define the authoritative epicentre, origin time, magnitude and depth for corridor construction?
2. Which finite-fault or tsunami-source reconstructions are candidates for regional bed forcing?
3. Which observations were used to infer a source and therefore cannot later be claimed as independent validation evidence?

4. Which data fields must Software receive to generate $\Delta b_m(x, y)$ and rupture-time functions reproducibly?

4.3 Terminology and Definitions

4.4 Literature Evidence

U.S. Geological Survey [1] is retained as the event-metadata source for the 2011 Great Tōhoku earthquake: origin time 2011-03-11 05:46:24 UTC, epicentral coordinates near $38.297°$N, $142.373°$E, depth 29.0 km and moment magnitude $M_{\mathrm{w}} = 9.1$. Source role: supports event identity, epicentre coordinate and corridor origin. Project conclusion: these metadata define $P_{\mathrm{e}} = (\lambda_{\mathrm{e}}, \phi_{\mathrm{e}})$ until a GIS-approved event-source point is selected. Limitation: event metadata alone do not define the spatial seabed displacement field.

U.S. Geological Survey [9] and Hayes [10] are retained for the USGS finite-fault source-data family and rapid Tōhoku source characterisation. Source role: supports candidate finite-fault model fields: subfault geometry, slip, rake, rupture time and rise time. Project conclusion: the USGS finite-fault family is a baseline source candidate for regional moving-bed forcing. Limitation: the finite-fault model must still be converted into hydrodynamic bed increments and tested against observation sets not used in the inversion.

Fujii et al. [11] and Wei et al. [12] remain retained source-reconstruction references. Source role: support and limit the tsunami-source calibration choices used in Tōhoku hindcasts. Project conclusion: these sources are comparison and provenance anchors when deciding whether the USGS finite-fault source or an alternative tsunami-inversion source better supports the selected Kamaishi and Sendai corridors. Limitation: any observation record used to tune a source shall be marked $\mathcal{D}_{\mathrm{cal}}$ in ‘RES-DAT-OBS‘.

4.5 Governing Relations and Quantitative Definitions

4.6 Required Data Variables and Units

The dynamic moving-bed regional source requires, for each source component or subfault m:

- geometry, location, depth, length and width;
- strike, dip and rake;
- slip or displacement amplitude;
- rupture timing and rise-time function $R_m(t)$;
- projected bed increment $\Delta b_m(x, y)$;
- provenance and uncertainty metadata.

4.7 Candidate Data Sources

The current source candidates are:

- USGS event metadata for origin time, magnitude, depth and epicentre [1];
- USGS finite-fault products and rapid finite-fault source characterisation [9, 10];
- Fujii–Satake source reconstruction [11];
- DART-constrained and nested-model source workflows described by Wei et al. [12].

The source-data table in ‘RES-DAT-CAS‘ records each candidate’s role, downstream deliverable and limitation.

4.8 Spatial and Temporal Coverage

Spatial coverage shall include the full rupture footprint and enough surrounding seafloor for interpolation of dynamic bed increments onto the regional mesh. Temporal coverage shall include the origin time, subfault rupture onset and rise-time function for every source component retained in the moving-bed source.

4.9 Coordinate and Datum Requirements

Raw source coordinates shall be retained in longitude/latitude and source-native depth convention. The solver-facing product shall be projected into the same horizontal metric frame as ‘RES-DAT-GEO‘ and shall use positive-upward bed increments compatible with Equation (4.1). Any conversion from fault slip to seabed displacement shall record the elastic model, coordinate frame, vertical sign convention and interpolation method.

4.10 Data Processing and Transformation

The hydrodynamic source data shall be delivered as bed increments compatible with:

$$b_i(t_s) = b_{0,i} + \sum_m \Delta b_{m,i} R_m(t_s) \tag{4.1}$$

where t_s is the relevant Runge–Kutta stage time. Static source initialisation may be retained as a verification or fallback case, but the regional model specification treats dynamic moving bathymetry as the selected mathematical source representation.

4.11 Uncertainty, Provenance and Licensing

The source manifest shall identify whether each observation used to create or tune a source is calibration, validation or diagnostic evidence. Finite-fault uncertainty shall be retained at least as source version, inversion method, data types used, subfault discretisation, rupture timing assumptions and published update date where available. Licence and access notes shall be recorded before any finite-fault data are stored in a project dataset bundle.

4.12 Data Acceptance Criteria

A source dataset is accepted for regional replay only when it supplies or can defensibly derive:

- event origin metadata and source version;
- subfault geometry, strike, dip, rake, slip and depth;
- rupture timing or a documented static-source fallback;
- projected bed increments $\Delta b_{m,i}$ on the regional mesh;
- calibration/validation-use classification for all observation records used in the source construction;
- provenance, licence and transformation metadata.

4.13 Candidate Approaches

4.14 Critical Comparison

4.15 Assumptions and Applicability

4.16 Limitations and Failure Conditions

4.17 Project-Specific Conclusions

The Tōhoku source decision is source-data finalisation, not solver implementation. The USGS event metadata define the current corridor origin and event identity. USGS finite-fault, Hayes rapid source, Fujii–Satake and DART-constrained source workflows remain source candidates until their projected bed increments are compared against the selected offshore, nearshore and terrestrial observation split.

4.18 Selected Approach and Rationale

Use a source-candidate manifest with USGS event metadata as the event identity baseline and USGS finite-fault/Fujii/DART-constrained reconstructions as candidate bed-forcing products. The selected hydrodynamic source shall be the candidate that best supports the Kamaishi and Sendai corridor validation split without reusing calibration records as independent validation.

4.19 Unresolved Decisions

The source-data variables required by the mathematical model are established. The exact final fault-source dataset, any comparison against independent source reconstructions, final rupture-time functions and the accepted source-to-bed projection remain open data-selection tasks.

4.20 Requirements and Handoffs

The selected source dataset shall be marked as calibration, validation-support or diagnostic evidence so ‘RES-VER‘ can distinguish source-fit records from independent validation records.

Software shall start a source-data ingestion path in parallel with the remaining research: source manifest, fault-model parser, coordinate projection, bed-increment rasterisation, structured source-output schema and reproducible moving-bed replay fields for the regional solver.

4.21 Deliverable Completion Record

5 RES-DAT-OBS — Observational and Validation Data

5.1 Purpose and Scope

This Deliverable now covers both event-validation observations and defence-specific evidence for documented structures exposed to the selected event. In addition to tsunami water-level and inundation observations, the catalogue shall include documented defence damage, survival, failure, pre-event geometry, post-event condition, current or reconstructed geometry, drawings, photographs, reconstruction records and local exposure evidence.

5.2 Research Questions

5.3 Terminology and Definitions

5.4 Literature Evidence

5.4.1 Mori et al. (2012)

Mori, Takahashi, and The 2011 Tohoku Earthquake Tsunami Joint Survey Group [2] documented the coordinated post-event survey of the 2011 Tohoku tsunami. The survey involved 299 researchers from 64 universities and institutes, covered approximately 2,000 km of the Japanese coastline, and contained 5,214 surveyed locations by the end of December 2011. The principal measured quantities were inundation height, run-up height, and inundation distance.

The paper is retained as the principal nationwide observational reference for validating maximum onshore water levels and inundation extent. It is not sufficient for validating velocity, discharge, momentum flux, impact pressure, or time-dependent structural loading, as the field dataset primarily records maximum water levels and landward limits.

5.4.2 Wei et al. (2013)

Wei et al. [12] demonstrated an observation hierarchy in which deep-ocean tsunami records constrained the source, while offshore GPS buoys, nearshore gauges, field-survey heights, and mapped inundation provided subsequent assessments of the regional and coastal model.

The paper is retained as evidence for separating source-calibration records from independent offshore and terrestrial validation data.

5.4.3 NOAA/NCEI DART and Tsunami Data

National Oceanic and Atmospheric Administration [3] documents NOAA/NCEI as the long-term archive for DART ocean bottom pressure data, including netCDF and CSV products. NOAA National Centers for Environmental Information [4] provides event-specific DART records for the March 11, 2011 Japan earthquake, including raw observations, computed tides and residual values. Source role: supports offshore time-series validation and source-calibration audit. Project-specific conclusion: DART records are mandatory candidates for regional propagation and arrival-time assessment, but any DART record used to infer a source shall be marked calibration evidence.

National Geophysical Data Center / World Data Service [13] documents the NCEI/WDS Global Historical Tsunami Database, including source-event and run-up records with observed maximum water heights, arrival or travel times and inundation information where available. Source role: supports event inventory, run-up record discovery and cross-checking against the Mori/JTS survey. Project-specific conclusion: the database is a screening and provenance source, not a replacement for the case-specific field-survey dataset selected for validation.

5.5 Observation Hierarchy

The benchmark observation set shall be organised as:

1. geodetic and seismic measurements for earthquake-source definition;
2. deep-ocean bottom-pressure records for source and regional propagation assessment;
3. DART residual water-level time series for arrival time, amplitude, phase and source-calibration provenance;
4. offshore GPS-wave-gauge records for nearshore transformation;
5. coastal, harbour and tide gauges for local waveform, arrival-time and maximum-water-height assessment;
6. satellite altimetry for spatial wave-front assessment;
7. surveyed inundation depth and run-up;
8. mapped inundation extent.

Each dataset shall record:

- spatial coordinates and reference datum;
- sampling interval;
- filtering and de-tiding procedure;
- measurement uncertainty;
- missing-data intervals;
- whether it was used for calibration, validation, or diagnosis;
- the model stage against which it is compared.

5.6 Calibration, Validation and Diagnostic Classification

Every observational record shall be classified as one of:

$$\mathcal{D} = \begin{cases} \mathcal{D}_{\mathrm{cal}}, & \text{used to infer or tune the source or model parameters} \\ \mathcal{D}_{\mathrm{val}}, & \text{withheld for independent assessment} \\ \mathcal{D}_{\mathrm{diag}}, & \text{used for qualitative or diagnostic comparison} \end{cases} \tag{5.1}$$

Agreement with $\mathcal{D}_{\mathrm{cal}}$ shall be described as a reconstruction fit rather than independent validation.

Where one observation record is divided temporally, the inversion window and withheld waveform interval shall be identified separately.

5.7 Governing Relations and Quantitative Definitions

5.8 Required Data Variables and Units

The observation record shall distinguish the following quantities:

- arrival time or travel time of the first observed wave;
- maximum water height at a gauge, buoy, survey or run-up location;
- tsunami height before or at shoreline arrival, in metres;
- inundation height at an onshore location, in metres above the selected sea-level datum;
- run-up height at the maximum landward extent, in metres above the selected sea-level datum;
- inundation distance from the shoreline, in kilometres or metres;
- observation latitude and longitude in decimal degrees;

- measurement method and instrument;
- surveyed trace type;
- tidal-correction method;
- survey date and, where available, survey time;
- observation-quality or reliability classification.

Tsunami height, inundation height, and run-up height shall not be treated as interchangeable validation variables.

5.9 Early and Late-Time Validation

The early coherent wave field shall be assessed deterministically using:

- arrival time;
- amplitude;
- waveform phase;
- dominant period;
- wave-front position.

Fixed-point time-series records and spatial wave-front observations shall be treated as complementary evidence rather than interchangeable validation products.

The later reflected and scattered wave field may additionally require:

- moving-window variance;
- spectral energy;
- waveform envelope;
- cumulative energy flux;
- cumulative force impulse;
- duration above a defined hazard threshold.

Statistical late-time agreement shall supplement rather than replace deterministic diagnosis. A persistent phase or amplitude discrepancy shall first be examined for errors in the source, bathymetry, boundaries, or numerical formulation.

5.10 Candidate Data Sources

Candidate defence-evidence sources include official survey reports, coastal engineering records, reconstruction drawings, site photographs, government or municipal infrastructure records, post-disaster damage surveys, peer-reviewed case studies and georeferenced imagery. Each source shall be classified by authority, date, provenance, licence or access constraint, and whether it supports geometry, exposure, damage, survival or reconstruction evidence.

Candidate event-observation sources are DART bottom-pressure records [3, 4], tide/coastal/harbour-gauge records identified through official archives, the NCEI/WDS Global Historical Tsunami Database [13], the Mori/JTS field survey [2], and the observation sets documented by Fine, Kulikov, and Cherniawsky [14] and Wei et al. [12]. Tide-gauge and harbour-gauge records remain a TODO–source-confirmation item until the specific station data and licence metadata for the two corridors are selected.

5.11 Spatial and Temporal Coverage

The survey extended approximately 2,000 km along the Japanese coastline. Maximum run-up greater than 10 m was reported along approximately 530 km of coast, while run-up greater than 20 m occurred along approximately 200 km. More than 1,000 observations were collected on the Sendai Plain, providing comparatively dense coverage for regional inundation assessment. A restricted zone of approximately 30 km around the Fukushima Daiichi Nuclear Power Plant was not surveyed. Observation density also varied substantially between regions and bays.

5.12 Coordinate and Datum Requirements

5.13 Data Processing and Transformation

Field measurements were corrected to remove the astronomical tide at the time of maximum tsunami elevation. Near-field corrections used the NAOTIDEJ astronomical tide database after many coastal gauges were destroyed. The time of maximum tsunami elevation was estimated using a nonlinear long-wave simulation based on the FS40v46 finite-fault source model. The resulting near-field observations are therefore not completely independent of the Fujii source reconstruction. This dependence shall be retained in the data-provenance record and considered when defining independent validation tests.

5.14 Uncertainty, Provenance and Licensing

The comparison between the astronomical tide database and surviving gauges produced a mean absolute tidal-elevation difference of approximately 6.0 cm. The complete relative tidal correction had a reported mean absolute difference of approximately 17.3 cm and a standard deviation of approximately 25.0 cm. These values represent only part of the total observation uncertainty. Additional uncertainty may arise from watermark interpretation, instrument choice, reference-datum conversion, survey timing, measurement density, and local spatial variability. A uniform pointwise uncertainty shall not be assigned without inspecting the metadata of the selected observations.

5.15 Data Acceptance Criteria

For G1 data feasibility, at least one defence shall have sufficiently reconstructable location, orientation, pre-event geometry, current or reconstructed geometry, material assumptions, foundation assumptions, observed event exposure, and damage, survival or reconstruction evidence.

Observational data selected for benchmark validation shall be accepted only when:

- the measured quantity is identified explicitly as tsunami height, inundation height, run-up height, or inundation distance;
- arrival time and maximum water height are retained when the source archive provides them;
- the horizontal coordinates and vertical datum are available;
- the tidal-correction procedure is documented;
- the observation is located within the resolved computational domain;
- the local topography and bathymetry are represented at a resolution consistent with the comparison;
- model output and observation refer to the same physical quantity;
- the observation is not used simultaneously for calibration and independent validation without explicit separation.

5.16 Candidate Approaches

5.17 Critical Comparison

5.18 Assumptions and Applicability

5.19 Limitations and Failure Conditions

5.20 Data Selection Conclusions

No single observation type is sufficient to validate the full model chain.

Deep-ocean records constrain source directivity and offshore propagation but do not establish accurate inundation. Inundation outlines constrain the flooded region but may conceal errors in local water level, velocity, and source amplitude. Satellite altimetry provides spatial wave-front information but has a lower signal-to-noise ratio than bottom-pressure records.

The benchmark shall therefore use complementary temporal, spatial, offshore, nearshore, and terrestrial observations.

5.21 Project-Specific Conclusions

The Mori et al. dataset shall be used to validate maximum inundation height, run-up height, and inundation extent within a selected regional case study. The complete nationwide dataset shall not be used as one undifferentiated objective function, as observation density, coastal regime, and data quality vary substantially between locations. The Sendai Plain is retained as a candidate regional inundation benchmark owing to its dense measurement coverage. Sanriku and Kamaishi Bay are retained as candidate local-transformation and barrier-study regions, subject to the acquisition of higher-resolution bathymetric, topographic, and dynamic observational data.

For the two active corridors, observation selection shall produce separate offshore, nearshore and terrestrial bundles. The Kamaishi bundle shall emphasise Sanriku/ria-coast transformation, bay/defence interaction and local barrier evidence. The Sendai coastal-plain bundle shall emphasise dense inundation, run-up and inland-extent observations. The bundles shall not be combined into one undifferentiated validation score.

5.22 Selected Approach and Rationale

5.23 Unresolved Decisions

5.24 Requirements and Handoffs

The numerical-verification workstream shall define separate error metrics for maximum inundation height, run-up height, and inundation distance. The mathematical- and numerical-model workstreams shall produce outputs that correspond directly to these observed quantities. Separate dynamic datasets shall be obtained for validating velocity, discharge, momentum flux, pressure, and impact loading. The present survey dataset shall not be used as sole validation evidence for those quantities.

Software shall preserve observation records in a manifest that records source archive, station or survey identifier, coordinates, vertical datum, time base, sampling interval, de-tiding/filtering procedure, calibration/validation/diagnostic classification, and linked model output variable.

5.25 Deliverable Completion Record

6 RES-DAT-SCN — Scenario and Severity Definition

6.1 Purpose and Scope

This Deliverable defines one primary historical tsunami event and controlled physical variations around that event for impact testing. Unrelated tsunami events shall not be added to the initial project scope.

The active scenario baseline is the 2011 Tōhoku event with two regional-to-local trajectories: epicentre to the Kamaishi region and epicentre to the Sendai coastal plain. Scenario definition owns the evidence-based corridor footprint; it does not define arbitrary drawing operations or solver implementation.

6.2 Research Questions

The scenario decision asks:

1. How are the two active regional-to-local corridors defined from event and target coordinates?
2. Which evidence determines corridor width, offshore length, inland length and sponge/relaxation margins?
3. Which scenario changes are active impact-test variations and which are deferred geometry or ML-generation tasks?

6.3 Terminology and Definitions

6.4 Literature Evidence

6.5 Governing Relations and Quantitative Definitions

6.5.1 Corridor Coordinate Basis

Base geographic coordinates are longitude/latitude. The event point is:

$$P_\mathrm{e} = (\lambda_\mathrm{e}, \phi_\mathrm{e}) \tag{6.1}$$

For a target coast point $P_\mathrm{t} = (\lambda_\mathrm{t}, \phi_\mathrm{t})$, a local metric displacement vector shall be constructed after projection or local tangent-plane conversion:

$$\Delta\mathbf{r} = [\Delta E,\ \Delta N]^\mathsf{T} \tag{6.2}$$

The signed rotation from the latitude/north axis to the corridor centreline is:

$$\theta = \mathrm{atan2}\,(\Delta E,\ \Delta N) \tag{6.3}$$

where positive angle is east of north under the selected projected coordinate convention. The rotated local basis is:

$$\hat{\mathbf{e}}_{x'} = [\sin\theta,\ \cos\theta]^\mathsf{T} \tag{6.4}$$
$$\hat{\mathbf{e}}_{y'} = [\cos\theta,\ -\sin\theta]^\mathsf{T} \tag{6.5}$$

For any projected point **r** measured from P_e, the corridor coordinates are:

$$x' = \Delta\mathbf{r} \cdot \hat{\mathbf{e}}_{x'} \tag{6.6}$$
$$y' = \Delta\mathbf{r} \cdot \hat{\mathbf{e}}_{y'} \tag{6.7}$$

Here x' is the corridor centreline/propagation direction and y' is the corridor-width direction. A constant-width baseline corridor is:

$$\mathcal{C} = \left\{ \mathbf{r} : -L_{\text{pre}} \le x' \le L_{\text{inland}}, \quad |y'| \le \frac{W}{2} \right\} \tag{6.8}$$

where W is corridor width, L_{pre} is offshore/upstream pre-impact length and L_{inland} is inland/downstream coverage. Sponge or relaxation margins are added outside $\mathcal{C}$ according to 'RES-DAT-GEO', 'RES-MOD-IBC' and 'RES-NUM-IBC'.

Decision trace: Source role: supports coordinate-transform and corridor-polygon generation for the data/model interface. Supporting citations: the event coordinate is anchored to U.S. Geological Survey [1]; corridor width and inland/offshore extent shall be supported by topobathymetric sources [5, 6, 8] and severe-impact observations [2, 13]. Limitation: the angles below use current coordinate proxies and must be recomputed from the final GIS-selected target coordinates.

6.6 Required Data Variables and Units

Required scenario variables are P_e, P_t, θ, W, L_{pre}, L_{inland}, upstream/downstream/lateral sponge widths, target-region identifier, observation-footprint polygon, defence-geometry footprint and source-data version.

6.7 Candidate Data Sources

6.8 Spatial and Temporal Coverage

The two active trajectories are:

T_K epicentre $\rightarrow$ Kamaishi region, using the current coordinate proxy $\theta_\mathrm{K} \approx -21.1°$.

T_S epicentre $\rightarrow$ Sendai coastal plain, using the current coordinate proxy $\theta_\mathrm{S} \approx -97.4°$.

The angles are not final GIS bearings. They shall be recomputed from the accepted target coordinates, selected projection and corridor origin before clipping or mesh generation.

6.9 Coordinate and Datum Requirements

6.10 Data Processing and Transformation

6.11 Uncertainty, Provenance and Licensing

6.12 Data Acceptance Criteria

A corridor definition is accepted only when:

- final P_e and P_t coordinates are recorded with source and CRS metadata;
- θ is recomputed from the final GIS coordinates;
- W, L_pre, L_inland and sponge widths are justified by source footprint, severe-impact observations, defence geometry, bathymetry/topography coverage and numerical relaxation requirements;
- observation and terrain datasets fully cover the corridor plus buffer;
- constant-width corridor adequacy is documented for each target.

6.13 Candidate Approaches

The active scenario approach is a validated event baseline plus bounded local variations in wave height, inundation depth, velocity, direction, duration and selected severity scaling. These variations support robustness and reinforcement screening within the event-conditioned domain.

The baseline corridor is constant-width. A narrowing or bay-following corridor is deferred unless GIS evidence shows that the narrowing represents true physical bay, coastline or bathymetric constraints rather than a drawing convenience.

6.14 Critical Comparison

The constant-width corridor is selected for the baseline because it creates a reproducible clipping and replay-boundary interface. It is limited where a ria bay, harbour mouth, river channel or coastal plain controls the flow in a way that a rectangular corridor would misrepresent. In those cases the physical constraint shall be encoded from GIS coastline/topography evidence, not manually narrowed for visual neatness.

6.15 Assumptions and Applicability

6.16 Limitations and Failure Conditions

6.17 Project-Specific Conclusions

The initial project is a single-event reconstruction and controlled impact-testing study, not a multi-event generalisation study.

The Tōhoku/local comparison is explicitly two-corridor: Kamaishi for a Sanriku/ria and defence-interaction case, and Sendai coastal plain for flat-plain inundation comparison. This pairing supports barrier-class comparison within one event while avoiding claims of cross-event generalisability.

6.18 Selected Approach and Rationale

Use two evidence-defined, constant-width baseline corridors: T_{K} and T_{S}. Corridor parameters are data products owned by RES-DAT, not solver constants. Source role: supports regional-to-local data extraction and replay-boundary generation. Downstream handoff: Software shall generate corridor polygons, clip raster/vector data, interpolate terrain, extract observation subsets and write replay-boundary definitions from the same manifest.

6.19 Unresolved Decisions

Open decisions are final GIS-selected target coordinates, final projection, corridor widths, upstream/downstream lengths, sponge widths, observation-footprint inclusion rules and whether a physically-constrained non-rectangular local boundary is required near Kamaishi Bay.

6.20 Requirements and Handoffs

Software shall start the corridor workflow in parallel with remaining research: coordinate-transform module, corridor-polygon generation, manifest-driven clipping, local inlet section extraction and replay-boundary metadata. ‘RES-VER-TST‘ shall use the same corridor parameters in interface-location and one-way-versus-coupled tests.

6.21 Deliverable Completion Record

7 RES-DAT-UNC — Data Quality, Uncertainty and Provenance

7.1 Purpose and Scope

This Deliverable defines the provenance, CRS, datum, resolution, licence and preprocessing uncertainty records required for all Tōhoku case-study data. It covers uncertainty traceability for source, topobathymetry, observations, corridor geometry and derived replay boundaries. It does not quantify final validation tolerances; those belong to ‘RES-VER-MET‘.

7.2 Research Questions

The uncertainty decision asks:

1. Which metadata are mandatory before a dataset can be ingested?
2. Which preprocessing operations change the meaning or error structure of the data?
3. Which uncertainty flags must travel into validation reports and local replay boundaries?

7.3 Terminology and Definitions

7.4 Literature Evidence

7.5 Governing Relations and Quantitative Definitions

Each data asset d shall carry a provenance record:

$$\mathcal{P}_d = \{s_d,\, c_d,\, \mathcal{R}_{h,d},\, \mathcal{R}_{v,d},\, \Delta_d,\, \ell_d,\, \mathcal{T}_d,\, \mathcal{U}_d\} \tag{7.1}$$

where s_d is source/provider, c_d is citation key, $\mathcal{R}_{h,d}$ is horizontal CRS, $\mathcal{R}_{v,d}$ is vertical datum, Δ_d is native resolution, ℓ_d is licence/usage note, $\mathcal{T}_d$ is the ordered preprocessing transform and $\mathcal{U}_d$ is the uncertainty flag set.

7.6 Required Data Variables and Units

Mandatory fields are dataset/source name, provider, citation key, URL, spatial coverage, temporal coverage, native resolution, horizontal CRS, vertical datum, licence/usage note, project role, downstream deliverable, preprocessing required and limitation.

7.7 Candidate Data Sources

7.8 Spatial and Temporal Coverage

7.9 Coordinate and Datum Requirements

No dataset shall be projected, merged or compared until its coordinate and datum state is known or explicitly marked unresolved. Where a source lacks confirmed CRS or datum metadata, the derived product shall be flagged:

$$a_d = \text{screening only} \tag{7.2}$$

until the metadata are resolved.

7.10 Data Processing and Transformation

Preprocessing operations requiring explicit audit are reprojection, vertical datum conversion, sign-convention conversion, tidal correction, de-tiding, filtering, smoothing, gap filling, clipping, raster/vector overlay, interpolation, mesh projection, source-to-bed projection and replay-boundary sampling.

7.11 Uncertainty, Provenance and Licensing

The uncertainty flag set shall include, where applicable:

- source-age or survey-date heterogeneity;
- native-resolution mismatch across merged products;
- unknown or mixed vertical datums;
- horizontal CRS conversion uncertainty;
- smoothing or interpolation of sharp bathymetric/defence features;
- tide correction or harmonic-analysis dependence;
- source-inversion dependence for observation products;
- licence restrictions preventing redistribution;
- missing station, survey or defence metadata.

Source role: supports validation traceability and software ingestion. Supporting dataset citations are recorded in ‘RES-DAT-CAS‘. Limitation: these flags identify uncertainty sources; they do not replace quantitative uncertainty propagation.

7.12 Data Acceptance Criteria

A data asset is accepted for validation or solver input only when its provenance record satisfies:

$$\mathcal{P}_d \supseteq \{s_d,\, c_d,\, \mathcal{R}_{h,d},\, \mathcal{R}_{v,d},\, \Delta_d,\, \ell_d,\, \mathcal{T}_d\} \tag{7.3}$$

Records missing any of these fields may be used only for exploratory screening unless the missing field is formally waived with a documented project risk.

7.13 Candidate Approaches

7.14 Critical Comparison

7.15 Assumptions and Applicability

7.16 Limitations and Failure Conditions

7.17 Project-Specific Conclusions

The Tōhoku corridor comparison is sensitive to datum, resolution and preprocessing differences. A single unqualified terrain raster, source file or observation table is therefore not a valid project input. The minimum accepted object is a data asset plus its provenance record $\mathcal{P}_d$.

7.18 Selected Approach and Rationale

Use a manifest-first data workflow. Every source and derived product is registered before use, and every transformation adds a new derived record rather than overwriting the source identity.

7.19 Unresolved Decisions

Open decisions are final licence acceptance for J-EGG500/GSI-derived products, exact vertical datum transformations, quantitative terrain uncertainty values, observation station filtering rules and the threshold for treating a missing metadata field as blocking rather than screening-only.

7.20 Requirements and Handoffs

Software shall implement a dataset manifest, immutable source records, derived-product records, coordinate-transform audit logs, HDF5 or equivalent structured output metadata and validation-report provenance exports. ‘RES-VER-MET‘ shall consume the same uncertainty flags when deciding whether a comparison is admissible.

7.21 Deliverable Completion Record

8 RES-DAT-CAS — Integrated Tsunami Case-Study Specifications

8.1 Purpose and Scope

This Deliverable replaces the previous mandatory primary-plus-secondary-event specification with one authoritative event-reconstruction case, one validated event baseline, bounded impact-testing variants, and documented defence-specific local cases.

The active integrated case is the 2011 Tōhoku event with two regional-to-local corridors: Kamaishi and Sendai coastal plain. The case supports impact modelling, local comparison and barrier-class comparison. ML, surrogate optimisation and generative geometry are deferred until the hydrodynamic data pipeline and validation hierarchy are accepted.

8.2 Research Questions

The integrated case-study decision asks:

1. Which data sources define the event, terrain, observations, corridors and optional sensitivities?
2. Which source is selected, fallback, comparison-only, optional or missing?
3. Which downstream deliverable and software asset consumes each source?
4. Which limitations prevent a source from being used as validation or solver input?

8.3 Terminology and Definitions

8.4 Literature Evidence

The integrated case combines official data products and peer-reviewed Tohoku evidence. Event identity is anchored to the USGS event record [1]. Candidate finite-fault and tsunami source products are retained from USGS finite-fault documentation, Hayes rapid source characterisation, Fujii–Satake and DART-constrained hindcast evidence [9–12]. Terrain sources are drawn from J-EGG500, GSI DEM, GEBCO_2026 and ETOPO 2022 [5–8]. Observation sources include NOAA/NCEI DART, the NCEI/WDS historical tsunami database, Mori/JTS and published observation syntheses [2, 3, 4, 13, 14]. ERA5 and JRA-55 are retained only as optional wind/reanalysis sensitivity sources, not baseline hydrodynamic forcing [15, 16].

8.5 Governing Relations and Quantitative Definitions

8.6 Required Data Variables and Units

The integrated case-study manifest shall include:

- event metadata, finite-fault source version and derived bed increments;
- terrain source tiles and derived mesh-projected bed fields;
- observation bundles classified as calibration, validation or diagnostic;
- corridor polygons for T_{K} and T_{S};
- local barrier-class definitions and terrain/geometry patches;
- replay-boundary histories for local 3D inlet forcing;
- provenance records $\mathcal{P}_d$ for every source and derived product.

8.7 Candidate Data Sources

Table 1. RES-DAT integrated data-source decision table for the Tōhoku corridor case

Dataset/source name	Provider	Citation key	URL	Coverage, resolution, CRS and vertical datum	Licence/usage note	Project role and downstream deliverable	Preprocessing required and limitation
J-EGG500	Japan Oceanographic Data Center	`jodcJegg500`	https://www.jodc.go.jp/jodcweb/JDOSS/infoJEGG.html	Spatial: Japan surrounding seas. Temporal: static compiled bathymetry with heterogeneous survey dates. Resolution: 500 m grid. CRS/datum: confirm from delivered metadata before ingest; bathymetric sign convention requires conversion to positive-upward *b*.	Usage/licence to be confirmed from JODC terms before redistribution.	Preferred Japan offshore bathymetry candidate for 'RES-DAT-GEO', 'RES-DAT-CAS', regional mesh and corridor terrain.	Reproject, convert depth sign/datum, clip corridors, interpolate to meshes. Limitation: smoothing and mixed data quality may suppress harbour-scale relief.
GEBCO_2026 Grid	GEBCO Bathymetric Compilation Group	`gebco2026grid`	https://www.gebco.net/data-products/gridded-bathymetry-data	Spatial: global ocean/land terrain. Temporal: 2026 release. Resolution: 15 arc-second grid. CRS: geographic grid. Vertical datum: treated as mean sea level, with shallow-water datum caveat.	Public-domain use with required attribution/disclaimer.	Selected outer-domain and first fallback topobathymetry for 'RES-DAT-GEO', regional domain and source-to-coast propagation.	Clip outer domain, compare with J-EGG500/GSI boundaries, resample to mesh. Limitation: not sufficient for local defence geometry or high-resolution inundation features.
NOAA ETOPO 2022	NOAA/NCEI	`noaa2022etopo`	https://www.ncei.noaa.gov/products/etopo-global-relief-model	Spatial: global relief. Temporal: 2022 release. Resolution: 15, 30 and 60 arc-second products. CRS/datum: confirm product metadata; elevation/bathymetry grid for global relief.	NOAA citation and DOI required; redistribution terms to be checked with source package.	Secondary global fallback and audit source for 'RES-DAT-GEO' and outer-domain sensitivity.	Clip/resample and compare against GEBCO. Limitation: fallback only for local corridors and not baseline local terrain.
GSI DEM	Geospatial Information Authority of Japan	`gsiDemDownload`	https://service.gsi.go.jp/kiban/app/help/	Spatial: Japan onshore. Temporal: source-age varies by DEM class. Resolution: 1 m, 5 m and 10 m mesh products. CRS/datum: confirm in JPGIS/GML metadata before ingest.	GSI usage terms and attribution required; redistribution status to be confirmed.	Preferred onshore topography for 'RES-DAT-GEO', run-up/inundation comparison and local terrain patches.	Convert JPGIS/GML, reproject, vertical-datum reconcile, clip corridors. Limitation: mixed DEM classes and coverage gaps require provenance flags.
USGS Tōhoku event record	U.S. Geological Survey	`usgs2011tohokuEvent`	https://earthquake.usgs.gov/earthquakes/eventpage/official20110311054624120_30/executive	Spatial: epicentre near 38.297°N, 142.373°E. Temporal: 2011-03-11 05:46:24 UTC. Resolution: event metadata. CRS/datum: geographic coordinates, depth metadata.	USGS public information; cite source record.	Event identity and corridor origin for 'RES-DAT-SRC' and 'RES-DAT-SCN'.	Store immutable event metadata and convert to projected corridor origin. Limitation: does not define seabed displacement.

Dataset/source name	Provider	Citation key	URL	Coverage, resolution, CRS and vertical datum	Licence/usage note	Project role and downstream deliverable	Preprocessing required and limitation
USGS finite-fault family and Hayes rapid source	U.S. Geological Survey	`usgsFiniteFaults`; `hayes2011rapid`	https://earthquake.usgs.gov/data/finitefault/	Spatial: rupture/subfault footprint. Temporal: event rupture history. Resolution: subfault dependent. CRS/datum: source-model metadata required.	Cite USGS record and article; data-package usage terms to be checked.	Candidate source baseline for 'RES-DAT-SRC' and moving-bed forcing.	Parse subfaults, project geometry, compute Δb_m, rasterise to regional mesh. Limitation: calibration observations must not be reused as independent validation.
Fujii–Satake Tōhoku source	Fujii et al.	`fujii2011tsunamisource`	DOI in bibliography	Spatial: 2011 Tōhoku source region. Temporal: event source reconstruction. Resolution: source-model dependent. CRS/datum: paper/model metadata required.	Journal citation required.	Comparison source and provenance anchor for 'RES-DAT-SRC' and observation de-tiding dependence.	Reconstruct source-to-bed projection if selected. Limitation: survey corrections in Mori et al. partly depend on FS40v46-style modelling.
NOAA/NCEI DART ocean bottom pressure	NOAA/NCEI	`noaa2005dart`; `noaa2011japanDart`	https://www.ncei.noaa.gov/products/natural-hazards/tsunamis-earthquakes-volcanoes/tsunamis/dart-ocean-bottom-pressure	Spatial: DART station points, Pacific/global network. Temporal: event time series and archive records. Resolution: includes 15-second bottom-pressure records where available. CRS/datum: station metadata and pressure-to-water-level conversion required.	Cite NOAA/NCEI; data access unrestricted through archive tools unless station-specific terms state otherwise.	Offshore arrival, phase, amplitude and source-calibration audit for 'RES-DAT-OBS' and 'RES-VER-SPEC'.	Download station records, de-tide/filter, classify calibration/validation. Limitation: DART records used for source inference are calibration, not independent validation.
NCEI/WDS Global Historical Tsunami Database	NOAA/NCEI/WDS	`noaaNceiHistoricalTsunami`	https://www.ncei.noaa.gov/products/natural-hazards/tsunamis-earthquakes-volcanoes/tsunamis/global-historical-data	Spatial: global tsunami source and run-up records. Temporal: historical-to-present database. Resolution: event/run-up record level. CRS/datum: per-record metadata; datum may be ambiguous.	DOI citation required; access unrestricted according to metadata.	Observation discovery, arrival-time and max-water-height screening for 'RES-DAT-OBS'.	Query Tōhoku records, cross-check against field survey. Limitation: heterogeneous historical provenance; not sole validation source.
Mori/JTS field survey	Mori, Takahashi and 2011 Tohoku Earthquake Tsunami Joint Survey Group	`mori2012nationwide`	DOI in bibliography	Spatial: approximately 2,000 km of Japanese coastline. Temporal: post-event 2011 survey. Resolution: surveyed point/extent records. CRS/datum: survey metadata and tide corrections required.	Journal citation required; raw-data access/licence to be confirmed.	Principal run-up, inundation-height and inundation-distance evidence for 'RES-DAT-OBS', Sendai and Sanriku/Kamaishi corridor validation.	Filter by corridor, datum-normalise, classify quality. Limitation: primarily maximum water-level and extent data, not velocity or pressure histories.

Dataset/source name	Provider	Citation key	URL	Coverage, resolution, CRS and vertical datum	Licence/usage note	Project role and downstream deliverable	Preprocessing required and limitation
Fine et al. observation/modelling synthesis	Fine et al.	fine2013japan	DOI in bibliography	Spatial: Japan 2011 tsunami observation/model context. Temporal: 2011 event. Resolution: paper-level synthesis. CRS/datum: underlying records must be obtained separately.	Journal citation required.	Supports observation hierarchy and regional propagation interpretation for 'RES-DAT-OBS'.	Use as evidence, not raw data. Limitation: cannot replace station/survey archives.
ERA5 and JRA-55	Copernicus Climate Change Service; Japan Meteorological Agency	copernicusEra5; jmaJra55	https://climate.copernicus.eu/climate-reanalysis; https://www.data.jma.go.jp/jra/html/JRA-55/index_en.html	Spatial: global atmospheric reanalysis. Temporal: multi-decadal products. Resolution: product dependent. CRS/datum: atmospheric grids.	Product-specific acknowledgement and terms required.	Optional wind/reanalysis sensitivity only; not baseline hydrodynamic forcing.	No baseline preprocessing. Limitation: not used unless a sensitivity question is opened and justified.
Kamaishi bay-mouth breakwater defence records	TODO–provider	TODO–missing-source	TODO	Spatial: Kamaishi Bay and defence footprint. Temporal: pre-2011, 2011 damage/response and reconstruction states. Resolution/CRS/datum: unresolved.	Licence unresolved.	Required for real defence recreation and wall-type reference geometry in 'RES-GEO' and local 'RES-MOD'/'RES-VER'.	Locate authoritative drawings, surveys, damage reports or peer-reviewed defence papers. Limitation: no citation is created until source metadata are confirmed.

8.8 Spatial and Temporal Coverage

The integrated case covers the 2011 Tōhoku source-to-coast event and two active regional-to-local corridor bundles:

1. T_{K}: epicentre to Kamaishi region, with Sanriku/ria, bay and defence-interaction emphasis;
2. T_{S}: epicentre to Sendai coastal plain, with coastal-plain inundation emphasis.

The local 3D replay interval shall include all waves that materially affect run-up, overtopping, transmitted/reflected energy and load metrics, not only the first crest.

8.9 Coordinate and Datum Requirements

All source coordinates are acquired as longitude/latitude and then transformed into the selected projected metric CRS for regional and local modelling. Vertical data shall be reconciled into the positive-upward bed convention used by 'RES-MOD' and 'RES-NUM'. Observation heights shall preserve their source datum and tide correction before any comparison to simulated η, run-up or inundation variables.

8.10 Data Processing and Transformation

The integrated data workflow is:

1. register source assets in the dataset manifest;
2. choose final P_{e}, Kamaishi target and Sendai target coordinates;
3. recompute θ_{K} and θ_{S};
4. generate corridor polygons and buffer/sponge polygons;
5. clip and reconcile terrain, source and observation datasets;
6. interpolate terrain and bed increments to regional mesh fields;
7. extract regional interface histories for local 3D replay;
8. write structured HDF5 or equivalent outputs with provenance.

8.11 Uncertainty, Provenance and Licensing

The integrated case shall not be accepted unless every source in Table 1 has a provenance record, licence/usage note and downstream role. Missing-source records are allowed only as explicit blockers or TODOs; they shall not be converted into fabricated citation keys.

8.12 Data Acceptance Criteria

An impact case is accepted only after event reconstruction validation and defence-model credibility are established. The case record shall identify the event baseline, local exposure, defence geometry version, material and foundation assumptions, numerical model fidelity, convergence state, physical validity state and uncertainty state.

For the current hydrodynamic baseline, an integrated case is accepted for software implementation planning when it contains the dataset manifest, coordinate transform, corridor polygons, clipped terrain, source candidate records, observation bundles and local replay-boundary format. This acceptance does not mean validation has been executed.

8.13 Candidate Approaches

8.14 Critical Comparison

8.15 Assumptions and Applicability

8.16 Limitations and Failure Conditions

8.17 Project-Specific Conclusions

The initial case-study specification is event-conditioned and defence-specific. Secondary events and cross-event holdouts are deferred until the single-event workflow is complete and reviewed.

The current integrated case is not a generic literature review. Each source is retained because it supports, limits or defers a project decision: terrain construction, source forcing, observation validation, corridor definition, optional sensitivity or missing defence geometry.

8.18 Selected Approach and Rationale

Use a manifest-driven Tōhoku case-study specification:

> USGS event identity and source candidates + J-EGG500/GSI/GEBCO/ETOPO terrain hierarchy + NOAA/NCEI and Mori/JTS observations + Kamaishi and Sendai corridor polygons + structured local replay-boundary outputs

ERA5 and JRA-55 are explicitly optional sensitivity sources rather than baseline forcing. Kamaishi defence records remain a missing-source blocker for real defence recreation.

8.19 Unresolved Decisions

Open decisions are final GIS target coordinates, final terrain product licence acceptance, final finite-fault/source selection, final calibration/validation split, tide-gauge station selection, Kamaishi defence-source acquisition, final corridor dimensions and final replay boundary schema.

8.20 Requirements and Handoffs

Data handling now begins in parallel with remaining research. Software shall start with a modular MVP containing:

- finite-volume data model and mesh/field containers;
- manifest-driven raster/vector I/O;
- coordinate-transform and corridor-polygon generation;
- bathymetry/topography interpolation and clipping;
- regional 2D scaffold and boundary framework;
- HDF5 or equivalent structured output schema;
- replay-boundary format for local 3D inlet forcing;
- local 3D scaffold that consumes replay histories without implementing the solver in this Research deliverable.

'RES-VER-SPEC' shall consume the same manifest when deciding which records are calibration, validation or diagnostic evidence.

8.21 Deliverable Completion Record

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